

Chapter 5: The Discrete-Time Fourier Transform

Fan Liu
Assistant Professor
Department of Electrical & Electronic Engineering

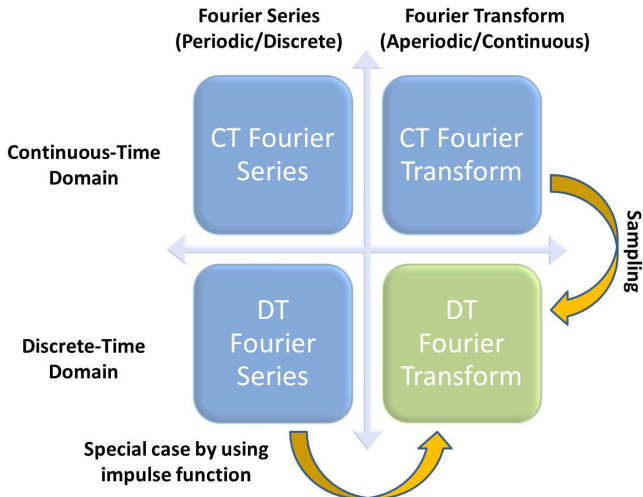
2022-Fall
Last Update on: Monday 7th November, 2022



Homework & Tutorial Problems

- Tutorial
 - ▶ Property of DTFT: 5.24
 - ▶ Difference equation of LTI systems: 5.36
- Homework
 - ▶ 5.23, 5.29, 5.33

A Big Picture



Outline

- Definition of discrete time Fourier transform (DTFT)
 - ▶ From Fourier series to Fourier transform
 - ▶ Convergence issue
- Periodic signals' DTFT
- Properties of DTFT
- Case study: Orthogonal frequency division multiplexing (OFDM)



Discrete-Time Fourier Transform

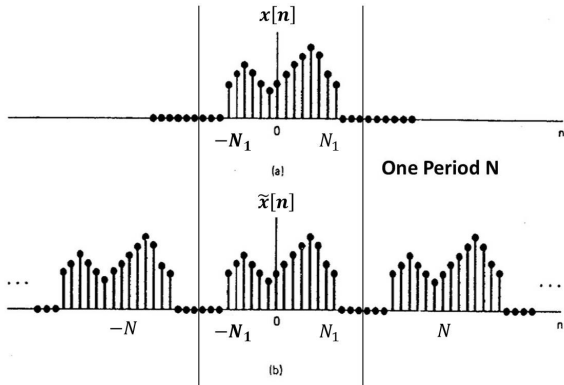
- Discrete-time Fourier series (for periodic signals)

$$\text{Synthesis Equation: } x[n] = \sum_{k=\langle N \rangle} a_k e^{jk(2\pi/N)n}$$

$$\text{Analysis Equation: } a_k = \frac{1}{N} \sum_{n=\langle N \rangle} x[n] e^{-jk(2\pi/N)n}$$

- Aperiodic signals can be treated as periodic signals with period $N \rightarrow \infty$
 - $x[n]$ must be like $\sum_k b_k e^{jk(2\pi/N)n}$ or $\int_{\omega} b(\omega) e^{j\omega n} d\omega$
 - b_k or $b(\omega)$ can be calculated from $x[n]$

Derivation (1/3)



- Original signal: $x[n]$
- Define new periodic signal with period N : $\tilde{x}[n]$, such that

$$\tilde{x}[n] = x[n], \quad n = -N/2, \dots, N/2 - 1$$

- Notice: when $N \rightarrow \infty$, $\tilde{x}[n]$ becomes $x[n]$

Derivation (2/3)

- Look at the Fourier series of $\tilde{x}[n]$:

$$a_k = \frac{1}{N} \sum_{n=-N/2}^{N/2+1} \tilde{x}[n] e^{-jk(2\pi/N)n} = \frac{1}{N} \sum_{n=-\infty}^{\infty} x[n] e^{-jk(2\pi/N)n}$$

- Define $X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n}$, so that $a_k = X(e^{jk\omega_0})/N$ ($\omega_0 = 2\pi/N$)
- Look at the synthesis equation:

$$\begin{aligned} \tilde{x}[n] &= \sum_{k=\langle N \rangle} a_k e^{jk(2\pi/N)n} = \frac{1}{N} \sum_{k=\langle N \rangle} X(e^{jk\omega_0}) e^{jk\omega_0 n} \\ &= \frac{1}{2\pi} \sum_{k=\langle N \rangle} X(e^{jk\omega_0}) e^{jk\omega_0 n} \omega_0 \end{aligned}$$

- When $N \rightarrow \infty$, $\omega_0 \rightarrow d\omega$, $\sum \rightarrow \int$, $k\omega_0 \rightarrow \omega$, $\tilde{x}[n] \rightarrow x[n]$, thus,

$$x[n] = \frac{1}{2\pi} \int_{2\pi} X(e^{j\omega}) e^{j\omega n} d\omega$$

Navigation icons: back, forward, search, etc.



Derivation (3/3)

- Therefore, we get the discrete-time Fourier transform pair

Discrete-Time Fourier Transform

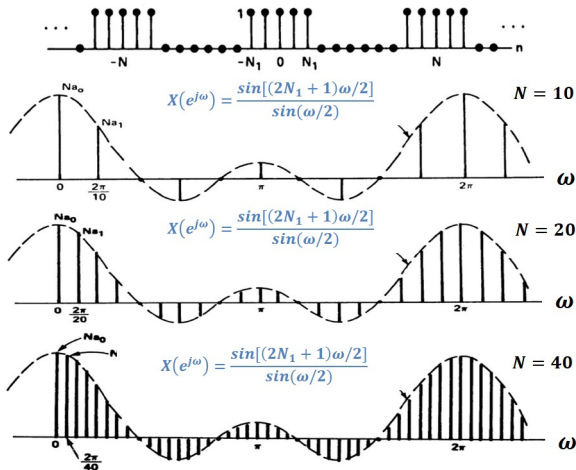
$$\text{Synthesis Equation: } x[n] = \frac{1}{2\pi} \int_{2\pi} X(e^{j\omega}) e^{j\omega n} d\omega$$

$$\text{Analysis Equation: } X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n}$$

- Observations
 - ▶ Continuous spectrum: Similar to CTFT
 - ▶ Periodic with period 2π : Different from CTFT
 - ▶ Low frequency: close to 0 and 2π ; high frequency: close to π

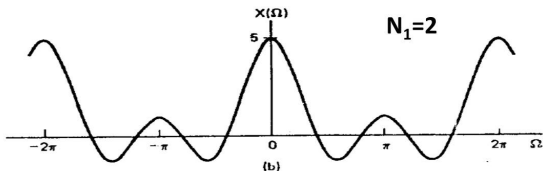
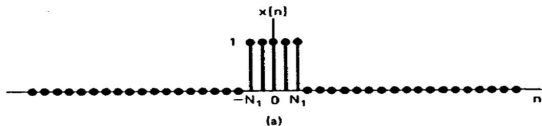
Example: From Periodic To Aperiodic

$$Na_k = \frac{\sin \frac{2\pi k \left(N_1 + \frac{1}{2}\right)}{N}}{\sin \left(\frac{\pi k}{N}\right)} = X(e^{j\frac{2\pi k}{N}})$$



Larger N , Denser Harmonic Components

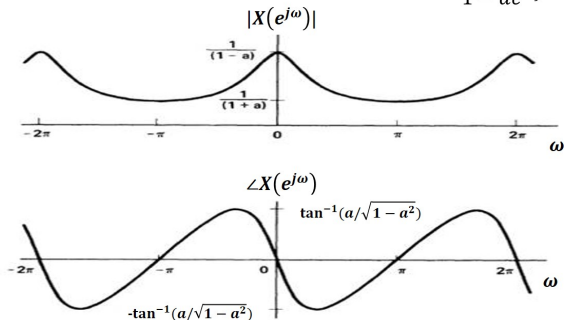
Fourier Transform Examples (1/2)



- $x[n] = 1$ ($n = -N_1, \dots, 0, \dots, N_1$)
- $X(e^{j\omega}) = \frac{\sin(\omega(N_1+1/2))}{\sin(\omega/2)}$
- Width of $x[n]$: $W_t = 2N_1 + 1$; width of $X(e^{j\omega})$: $W_f = \frac{4\pi}{2N_1+1}$
- $W_t \times W_f = 4\pi$, which is a constant

Fourier Transform Examples (2/2)

$$x[n] = a^n u[n] \quad 0 < a < 1 \quad \Leftrightarrow \quad X(e^{j\omega}) = \frac{1}{1 - ae^{-j\omega}}$$



- See textbook, Example 5.1
- What's the shape of magnitude when $a \rightarrow 1$ or $a \rightarrow 0$?

Convergence Issue of Analysis Equation

Sufficient Condition of Convergence

The analysis equation $X(e^{j\omega}) = \sum_{-\infty}^{\infty} x[n]e^{-j\omega n}$ will converge either if $x[n]$ is absolutely summable or if the sequence has finite energy, thus,

$$\sum_{-\infty}^{\infty} |x[n]| < \infty \text{ or } \sum_{-\infty}^{\infty} |x[n]|^2 < \infty$$

• Do the following signals have Fourier transform:

- ▶ $a^n u[n]$ ($0 < a < 1$)
- ▶ $\delta[n]$
- ▶ $u[n]$
- ▶ $e^{j\frac{2}{5}\pi n}$, $\cos(\frac{2}{5}\pi n)$
- ▶ $a^n u[n]$ ($a > 1$)

Can Periodic Signals Have DTFT?

- Definition of DTFT:

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n}$$

- Justification of divergence:

- ▶ Let $\omega = 2k\pi$, we have $X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]$
- ▶ Since $x[n]$ is periodic, the summation $\sum_{n=-\infty}^{\infty} x[n]$ will never converge unless $x[n] = 0$

- **Conclusion:** Most of periodic signals do NOT have DTFT according to the definition
- However, it's of significant engineering importance to extend Fourier transform to periodic signals

DTFT with Periodic Signals (1/2)

Fourier Transform of $e^{j\omega n}$

The following transform pair is actually NOT rigorously defined:

$$x[n] = e^{j\omega_0 n} \longleftrightarrow X(e^{j\omega}) = \sum_{l=-\infty}^{\infty} 2\pi\delta(\omega - \omega_0 - 2\pi l)$$

- **Synthesis:**

$$\frac{1}{2\pi} \int_{2\pi} X(e^{j\omega}) e^{j\omega n} d\omega = \frac{1}{2\pi} \int_{2\pi} \sum_{l=-\infty}^{\infty} 2\pi\delta(\omega - \omega_0 - 2\pi l) e^{j\omega n} d\omega = e^{j\omega_0 n}$$

- **Analysis:**

$$\sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n} = \sum_{n=-\infty}^{\infty} e^{j(\omega_0 - \omega)n} \quad \text{converge??}$$

DTFT with Periodic Signals (2/2)

- According to the Fourier series, for a periodic signal with period N :

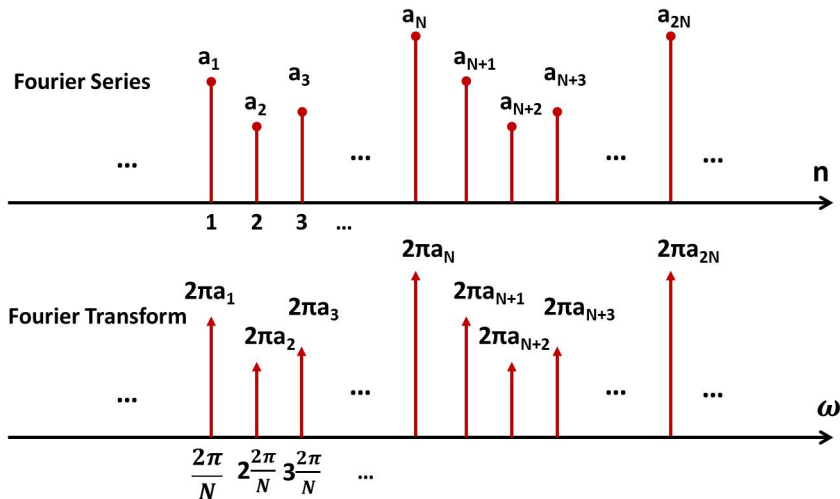
$$x[n] = a_0 + a_1 e^{j(2\pi/N)n} + \dots + a_k e^{jk(2\pi/N)n} + \dots + a_{N-1} e^{j(N-1)(2\pi/N)n}$$

- $e^{jk(2\pi/N)n} \longleftrightarrow \sum_{l=-\infty}^{\infty} 2\pi \delta(\omega - k(2\pi/N) - 2\pi l)$
- Then, due to the linearity of Fourier transform

$$\begin{aligned} \mathcal{F}\{x[n]\} &= \sum_{k=0}^{N-1} a_k \mathcal{F}\{e^{jk(2\pi/N)n}\} = \sum_{k=0}^{N-1} a_k \sum_{l=-\infty}^{\infty} 2\pi \delta(\omega - k(2\pi/N) - 2\pi l) \\ &= \sum_{l=-\infty}^{\infty} \sum_{k=0}^{N-1} a_k 2\pi \delta(\omega - k(2\pi/N) - 2\pi l) \end{aligned}$$

- Fourier transform of a periodic signal is a periodic sequence of impulses
 - What's the period? How many impulses within one period?

Fourier Series v.s. Fourier Transform



Example: Discrete-Time Impulse Chain

- What's the Fourier transform of $x[n] = \sum_{k=-\infty}^{+\infty} \delta[n - kN]$?
- First of all, we calculate the Fourier series:

$$\begin{aligned}a_k &= \frac{1}{N} \sum_{n=\langle N \rangle} x[n] e^{-jk \frac{2\pi}{N} n} \\&= \frac{1}{N} \sum_{n=\langle N \rangle} \sum_{k=-\infty}^{+\infty} \delta[n - kN] e^{-jk \frac{2\pi}{N} n} \\&= \frac{1}{N} \sum_{n=\langle N \rangle} \delta[n] e^{-jk \frac{2\pi}{N} n} = \frac{1}{N}\end{aligned}$$

- Then, we have

$$X(e^{j\omega}) = \sum_{l=-\infty}^{\infty} \sum_{k=0}^{N-1} \frac{2\pi}{N} \delta(\omega - k(2\pi/N) - 2\pi l) = \frac{2\pi}{N} \sum_{k=-\infty}^{\infty} \delta(\omega - \frac{2\pi k}{N})$$

Periodicity, Linearity and Shifting

- Periodicity

$$X(e^{j(\omega+2\pi)}) = X(e^{j\omega})$$

- ▶ How about CTFT? Why?

- Linearity

$$ax_1[n] + bx_2[n] \longleftrightarrow aX_1(e^{j\omega}) + bX_2(e^{j\omega})$$

- Time Shifting and Frequency Shifting

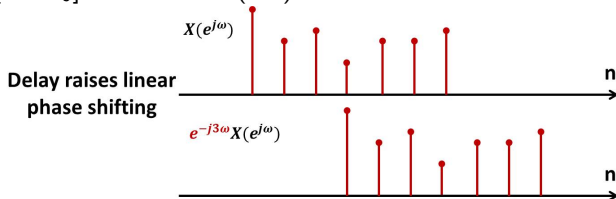
$$\begin{aligned}x[n - n_0] &\longleftrightarrow e^{-j\omega n_0} X(e^{j\omega}) \\e^{j\omega_0 n} x[n] &\longleftrightarrow X(e^{j(\omega - \omega_0)})\end{aligned}$$

- ▶ What's the physical meaning?

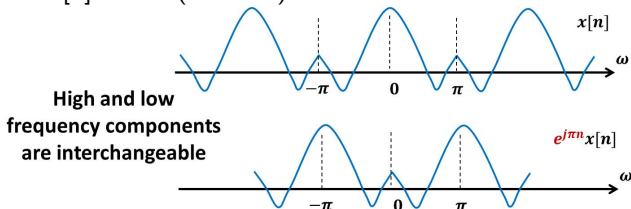


Illustration on Shifting

- $x[n - n_0] \longleftrightarrow e^{-j\omega n_0} X(e^{j\omega})$



- $e^{j\omega_0 n} x[n] \longleftrightarrow X(e^{j(\omega - \omega_0)})$



Conjugation, Differencing and Accumulation

- Conjugation

$$x^*[n] \longleftrightarrow X^*(e^{-j\omega})$$

- ▶ $X(e^{j\omega}) = X^*(e^{-j\omega}) \Leftrightarrow x[n]$ is real
- ▶ If $x[n]$ is real, then $\Re\{X(e^{j\omega})\}$ is even, $\Im\{X(e^{j\omega})\}$ is odd

- Differencing

$$x[n] - x[n-1] \longleftrightarrow (1 - e^{-j\omega})X(e^{j\omega})$$

- ▶ High-pass or low-pass?

- Accumulation

$$\sum_{m=-\infty}^n x[m] \longleftrightarrow \frac{1}{1 - e^{-j\omega}} X(e^{j\omega}) + \pi X(e^{j0}) \sum_{k=-\infty}^{\infty} \delta(\omega - 2\pi k)$$

- ▶ How to derive it via differencing?

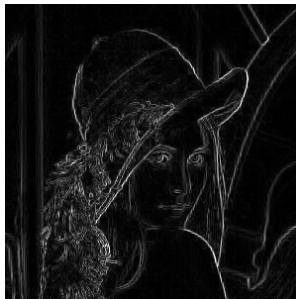
Problem 5.37

Problem

Let $X(e^{j\omega})$ be the Fourier transform of $x[n]$. Derive expressions in terms of $X(e^{j\omega})$ for the Fourier transforms of the following signals.

- $\text{Re}\{x[n]\}$
- $x^*[-n]$
- $\text{Ev}\{x[n]\}$

Effect of Differencing



- $J(i,j) = |M(i,j) - M(i+1,j+1)| + |M(i+1,j) - M(i,j+1)|$

Fourier Transform of $u[n]$

- How to derive the Fourier transform of $u[n]$?
- Option 1: From definition of Fourier transform

$$\mathcal{F}\{u[n]\} = \sum_{n=-\infty}^{+\infty} u[n]e^{-j\omega n} = \sum_{n=0}^{+\infty} e^{-j\omega n} \quad \text{Converge??}$$

- Option 2: Since $u[n] = \sum_{m=-\infty}^n \delta[m]$, according to the property of accumulation

$$\mathcal{F}\{u[n]\} = \frac{1}{1 - e^{-j\omega}} + \pi \sum_{k=-\infty}^{\infty} \delta(\omega - 2\pi k)$$

- Observation: Fourier transform of $u[n]$ does not exist according to the definition; however, it can be expressed in terms of $\delta(\cdot)$

Time Reversal and Expansion

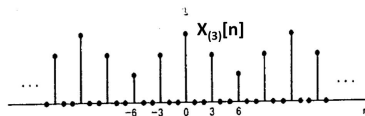
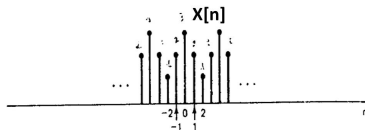
- Time Reversal

$$x[-n] \longleftrightarrow X(e^{-j\omega})$$

- Time Expansion

► Define $x_{(k)}[n] = \begin{cases} x[n/k], & \text{if } n \text{ is a multiple of } k \\ 0, & \text{Otherwise} \end{cases}$, then

$$x_{(k)}[n] \longleftrightarrow X(e^{jk\omega}) \quad (1)$$



Differentiation and Parseval

- Differentiation in Frequency

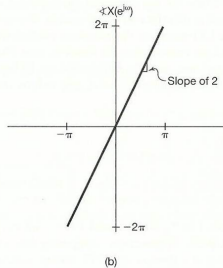
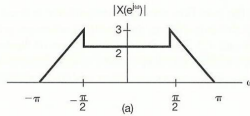
$$nx[n] \longleftrightarrow j \frac{dX(e^{j\omega})}{d\omega}$$

- Parseval's Relation

$$\sum_{n=-\infty}^{+\infty} |x[n]|^2 = \frac{1}{2\pi} \int_{2\pi} |X(e^{j\omega})|^2 d\omega$$

- ▶ Energy density spectrum — $\frac{|X(e^{j\omega})|^2}{2\pi}$

Example



- See textbook, Example 5.10
- Spectrum within $[-\pi, \pi]$
- Is it periodic, real, even, of finite energy?

Convolution Property & LTI Systems (1/2)

- Let $h[n]$ be the **impulse response** of certain LTI system
- The output $y[n]$ of input $x[n]$ is given by $y[n] = x[n] * h[n]$
- For input signal $x[n] = e^{j\omega n}$,

$$y[n] = \sum_{k=-\infty}^{\infty} h[k]e^{j\omega(n-k)} = e^{j\omega n} \underbrace{\sum_{k=-\infty}^{\infty} h[k]e^{-j\omega k}}_{\text{Frequency Response } H(e^{j\omega})}$$

- For periodic input signal $x[n] = \sum_{k=\langle N \rangle} a_k e^{jk(\frac{2\pi}{N})n}$:

$$y[n] = \sum_{k=\langle N \rangle} a_k H(e^{jk(\frac{2\pi}{N})}) e^{jk(\frac{2\pi}{N})n}$$

$$b_k = a_k H(e^{jk(\frac{2\pi}{N})})$$

Convolution Property & LTI Systems (2/2)

Convolution Property

If $y[n] = x_1[n] * x_2[n]$, then

$$Y(e^{j\omega}) = X_1(e^{j\omega})X_2(e^{j\omega})$$

- For general input signal $x[n]$:

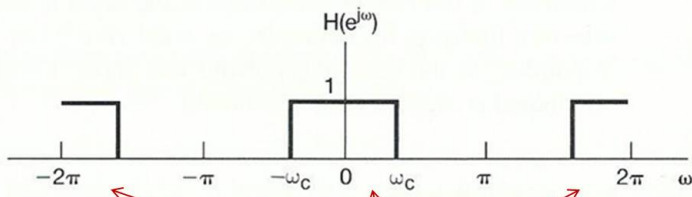
$$y[n] = x[n] * h[n] \longleftrightarrow Y(e^{j\omega}) = X(e^{j\omega})H(e^{j\omega})$$

- Observation:** It's easier to evaluate LTI systems in frequency domain
- Drawback:** Not every LTI system has frequency response
 - $h[n] = a^n u[n]$ ($a > 1$)
 - Stable LTI system has frequency response, because

$$\sum_{n=-\infty}^{\infty} |h[n]| < \infty$$

Example: Ideal Low-Pass Filter (1/2)

- What's ideal low-pass filter?
- Perfectly maintain the low-frequency component
- Perfectly cancel the high-frequency component

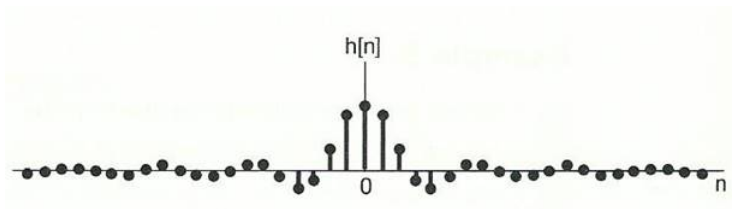


Repeat with period 2π

Example: Ideal Low-Pass Filter (2/2)

- Impulse response:

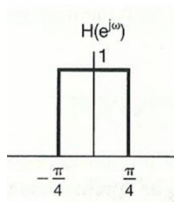
$$h[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} H(e^{j\omega}) e^{j\omega n} d\omega = \frac{1}{2\pi} \int_{-\omega_c}^{\omega_c} e^{j\omega n} d\omega = \frac{\sin \omega_c n}{\pi n}$$



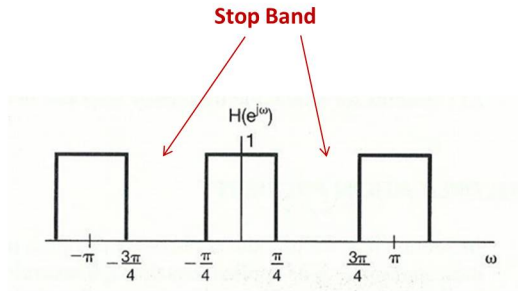
- Pros: no distortion in frequency domain
- Cons: non-causal
- See textbook, Example 5.12

Example: Band-Stop Filter (1/2)

- Given low-pass filter $H_{lp}(e^{j\omega})$, how to generate band-stop effect from low-pass filter?



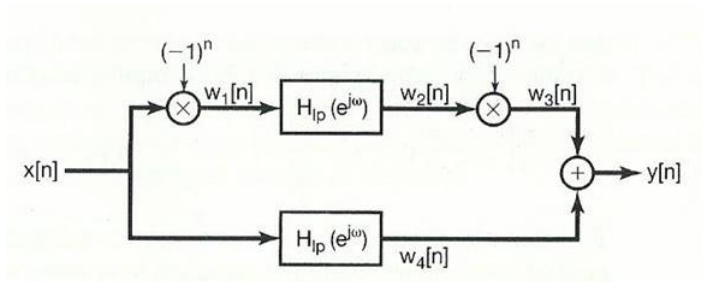
Low-Pass Filter



Band-Stop Filter

Example: Band-Stop Filter (2/2)

- Two branches: low-pass + high-pass



- $(-1)^n = e^{j\pi n} \Rightarrow W_1(e^{j\omega}) = X(e^{j(\omega-\pi)})$
 $\Rightarrow W_2(e^{j\omega}) = H_{lp}(e^{j\omega})X(e^{j(\omega-\pi)})$
 $\Rightarrow W_3(e^{j\omega}) = H_{lp}(e^{j(\omega-\pi)})X(e^{j(\omega-2\pi)}) = H_{lp}(e^{j(\omega-\pi)})X(e^{j\omega})$
- See textbook, Example 5.14

Multiplication Property

Multiplication Property

Let $y[n] = x_1[n]x_2[n]$, then

$$Y(e^{j\omega}) = \frac{1}{2\pi} \int_{2\pi} X_1(e^{j\theta}) X_2(e^{j(\omega-\theta)}) d\theta,$$

which is *periodic convolution* of $X_1(e^{j\omega})$ and $X_2(e^{j\omega})$

- Comparison: Multiplication of periodic signals

$$x_1[n] \longleftrightarrow a_k \quad \text{and} \quad x_2[n] \longleftrightarrow b_k$$

$$\Rightarrow x_1[n]x_2[n] \longleftrightarrow \sum_{k=\langle N \rangle} a_k b_{n-k} \quad \text{discrete-time periodic convolution}$$

Duality in DTFS

- Analysis and synthesis equations of DTFS share a similar form:

$$\begin{aligned}x[n] &= \sum_{k=\langle N \rangle} a_k e^{jk(2\pi/N)n} \\a_k &= \frac{1}{N} \sum_{k=\langle N \rangle} x[n] e^{-jk(2\pi/N)n}\end{aligned}\tag{2}$$

- Every of DTFS has a dual (See Table 3.2)
 - ▶ Time shift v.s. Frequency shift
 - ▶ Time multiplication v.s. Frequency multiplication
- If $g[n] \longleftrightarrow f[k]$, then $f[n] \longleftrightarrow \frac{1}{N}g[-k]$

Duality Between CTFS and DTFT

- DTFT transform pair is given by

$$\begin{aligned}x[n] &= \frac{1}{2\pi} \int_{2\pi} X(e^{j\omega}) e^{j\omega n} d\omega \\X(e^{j\omega}) &= \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n}\end{aligned}$$

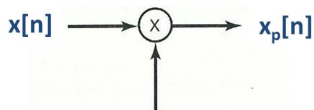
- CTFS transform pair is given by

$$\begin{aligned}x(t) &= \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t} \\a_k &= \frac{1}{T} \int_T x(t) e^{-jk\omega_0 t} dt\end{aligned}$$

- Frequency Domain of DTFT $\Leftrightarrow$ Time Domain of CTFS
- Frequency Domain of CTFS $\Leftrightarrow$ Time Domain of DTFT
- Discrete-Time Low-pass Filter v.s. Continuous-Time Square Wave

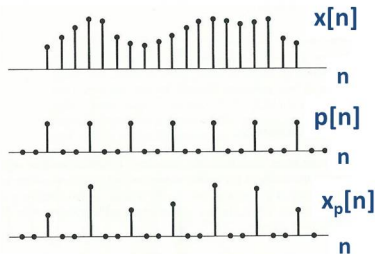
Example: Sampling on Discrete-Time Signals

- In digital signal processing, we may sample the discrete-time signals
- Example: image, audio, video compression
- Its mathematical model is give below:



$$p[n] = \sum_{k=-\infty}^{+\infty} \delta[n - kN]$$

$$x_p[n] = \sum_{k=-\infty}^{+\infty} x[kN] \delta[n - kN]$$



Review: Discrete-Time Impulse Chain

- What's the Fourier transform of $x[n] = \sum_{k=-\infty}^{+\infty} \delta[n - kN]$?
- First of all, we calculate the Fourier series:

$$\begin{aligned}a_k &= \frac{1}{N} \sum_{n=\langle N \rangle} x[n] e^{-jk \frac{2\pi}{N} n} \\&= \frac{1}{N} \sum_{n=\langle N \rangle} \sum_{k=-\infty}^{+\infty} \delta[n - kN] e^{-jk \frac{2\pi}{N} n} \\&= \frac{1}{N} \sum_{n=\langle N \rangle} \delta[n] e^{-jk \frac{2\pi}{N} n} = \frac{1}{N}\end{aligned}$$

- Then, we have

$$X(e^{j\omega}) = \sum_{l=-\infty}^{\infty} \sum_{k=0}^{N-1} \frac{2\pi}{N} \delta(\omega - k(2\pi/N) - 2\pi l) = \frac{2\pi}{N} \sum_{k=-\infty}^{\infty} \delta(\omega - \frac{2\pi k}{N})$$

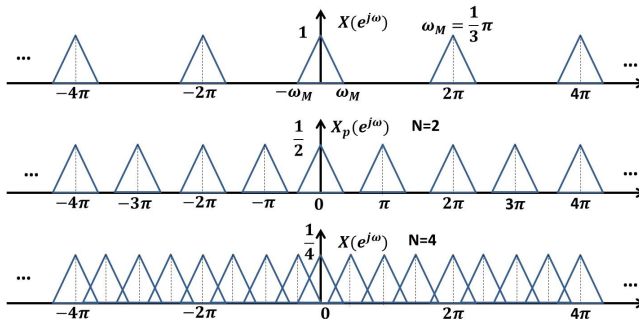
- Time domain period $\times$ Frequency domain period = ?



Example: Frequency Analysis

$$P(e^{j\omega}) = \frac{2\pi}{N} \sum_{k=-\infty}^{\infty} \delta(\omega - k\omega_s) \quad \text{where } \omega_s = 2\pi/N$$

$$X_p(e^{j\omega}) = \frac{1}{2\pi} \int_{2\pi} P(e^{j\theta}) X(e^{j(\omega-\theta)}) d\theta = \frac{1}{N} \sum_{k=0}^{N-1} X(e^{j(\omega-k\omega_s)})$$



Summary of Duality

	Continuous time		Discrete time	
	Time domain	Frequency domain	Time domain	Frequency domain
Fourier Series	$x(t) = \sum_{k=-\infty}^{+\infty} a_k e^{jk\omega_0 t}$ continuous time periodic in time	$a_k = \frac{1}{T_0} \int_{T_0} x(t) e^{-jk\omega_0 t} dt$ discrete frequency aperiodic in frequency	$x[n] = \sum_{k=\langle N \rangle} a_k e^{jk(2\pi/N)n}$ discrete time periodic in time	$a_k = \frac{1}{N} \sum_{k=\langle N \rangle} x[n] e^{-jk(2\pi/N)n}$ discrete frequency periodic in frequency
Fourier Transform	$x(t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} X(j\omega) e^{j\omega t} d\omega$ continuous time aperiodic in time	$X(j\omega) = \int_{-\infty}^{+\infty} x(t) e^{-j\omega t} dt$ continuous frequency aperiodic in frequency	$x[n] = \frac{1}{2\pi} \int_{2\pi} X(e^{j\omega}) e^{j\omega n} d\omega$ discrete time aperiodic in time	$X(e^{j\omega}) = \sum_{n=-\infty}^{+\infty} x[n] e^{-j\omega n}$ continuous frequency periodic in frequency

duality

duality

duality

- See textbook, Table 5.3



LTI by Difference Equation

- A number of DT LTI systems can be written as the following **linear constant-coefficient difference equation**

$$\sum_{k=0}^N a_k y[n-k] = \sum_{k=0}^M b_k x[n-k]$$

What's the frequency response?

- Taking Fourier transform on both side, we have

$$\begin{aligned} \sum_{k=0}^N a_k e^{-jk\omega} Y(e^{j\omega}) &= \sum_{k=0}^M b_k e^{-jk\omega} X(e^{j\omega}) \\ \Rightarrow H(e^{j\omega}) &= \frac{Y(e^{j\omega})}{X(e^{j\omega})} = \frac{\sum_{k=0}^M b_k e^{-jk\omega}}{\sum_{k=0}^N a_k e^{-jk\omega}} \end{aligned}$$

What's the impulse response?

Example: Difference Equation

Please calculate the frequency and impulse response of the following LTI systems

$$y[n] - \frac{3}{4}y[n-1] + \frac{1}{8}y[n-2] = 2x[n].$$

Solution

According to the last slide,

$$H(e^{j\omega}) = \frac{2}{1 - \frac{3}{4}e^{-j\omega} + \frac{1}{8}e^{-2j\omega}}.$$

Moreover, by partial fraction expansion

$$H(e^{j\omega}) = \frac{4}{1 - \frac{1}{2}e^{-j\omega}} - \frac{2}{1 - \frac{1}{4}e^{-j\omega}} \Rightarrow h[n] = 4\left(\frac{1}{2}\right)^n u[n] - 2\left(\frac{1}{4}\right)^n u[n]$$

Tutorial on Orthogonal Frequency Division Multiplexing (OFDM)



Discrete Fourier Transform (DFT)

- In practice, there is a huge demand on processing **finite duration signals**
- Given a finite duration signal $\{x[0], x[1], \dots, x[N-1]\}$, its Fourier transform is

$$X(e^{j\omega}) = \sum_{n=0}^{N-1} x[n]e^{-j\omega n}$$

- **Drawback:** The spectrum of DTFT is continuous $\Rightarrow$ Cannot be handle by computer.
- **Discrete Fourier Transfrom (DFT)** is developed for digital processing of finite duration signals

$$\tilde{X}[k] = \frac{1}{N} \sum_{n=0}^{N-1} x[n]e^{-jk(2\pi/N)n} \quad k = < N >$$

- DFT: frequency sampling of DTFT

$$\tilde{X}[k] = \frac{1}{N} X(e^{j2k\pi/N}) \quad k = < N >$$

Navigation icons: back, forward, search, etc.



Inverse DFT

- Equation system of DFT:

$$\begin{pmatrix} \tilde{X}[0] \\ \tilde{X}[1] \\ \vdots \\ \tilde{X}[N-1] \end{pmatrix} = \underbrace{\frac{1}{N} \begin{pmatrix} e^{-j0} & e^{-j0} & \dots & e^{-j0} \\ e^{-j0} & e^{-j2\pi/N} & \dots & e^{-j2(N-1)\pi/N} \\ \vdots & \vdots & \ddots & \vdots \\ e^{-j0} & e^{-j2(N-1)\pi/N} & \dots & e^{-j2(N-1)(N-1)\pi/N} \end{pmatrix}}_F \begin{pmatrix} x[0] \\ x[1] \\ \vdots \\ x[N-1] \end{pmatrix}$$

- Transform matrix F is full rank.
- Observation:** $\{\tilde{X}[k] | k = 0, \dots, N-1\}$ maintains all the information of $\{x[0], x[1], \dots, x[N-1]\}$
- Inverse DFT is feasible:

$$x[n] = \sum_{k=0}^{N-1} \tilde{X}[k] e^{jk(2\pi/N)n} \quad n = 0, 1, \dots, N-1$$

DFT and DTFS

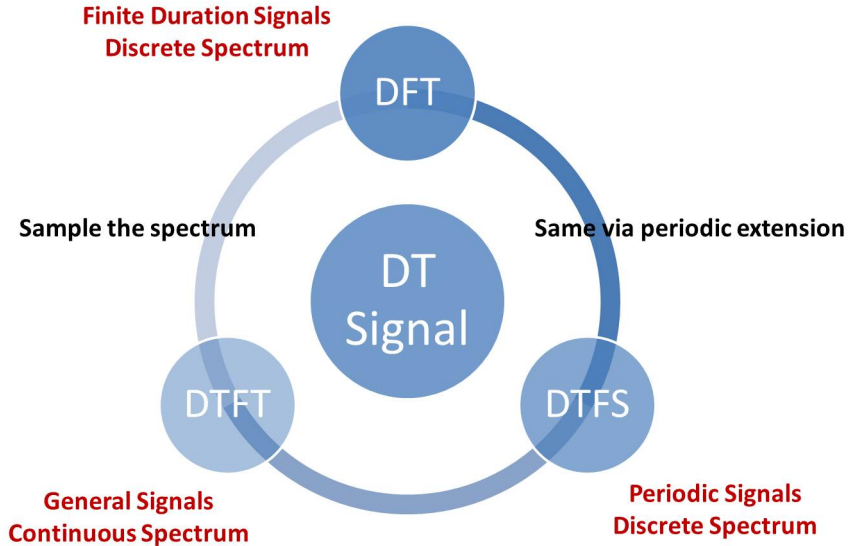
- DFT is for finite duration signals; DTFS is for periodic signals
- Define $\tilde{x}[n]$ as the periodic extension of $x[n]$: Repeat $x[n]$ with period N
- Fourier series of $\tilde{x}[n]$:

$$\frac{1}{N} \sum_{n=0}^{N-1} \tilde{x}[n] e^{-jk(2\pi/N)n} = \frac{1}{N} \sum_{n=0}^{N-1} x[n] e^{-jk(2\pi/N)n} \Rightarrow \text{DFT of } x[n]$$

- DFT of a finite-duration signal = DTFS of its periodic extension
- Reference on DFT:
 - ▶ Textbook: Problem 5.53, 5.54
 - ▶ http://en.wikipedia.org/wiki/Discrete_Fourier_transform



Comparison



Periodic Convolution of Finite Duration Signals

Periodic Convolution

Let x and y be two finite duration signals with duration N , $\tilde{x}$ and $\tilde{y}$ be the associated periodic extension, then the periodic convolution of finite duration signals is defined as

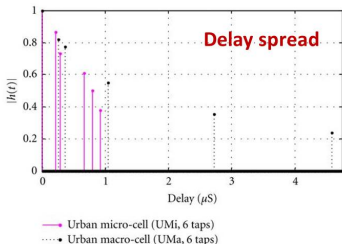
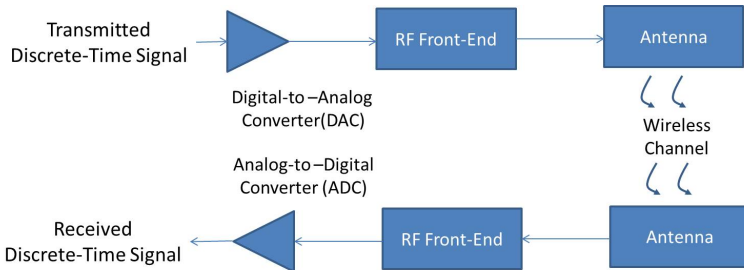
$$x[n] \circledast y[n] := \tilde{x}[n] \circledast \tilde{y}[n] = \sum_{k=\langle N \rangle} \tilde{x}[k] \tilde{y}[n - k] \quad n = 0, 1, 2, \dots, N - 1$$

Convolution Property of DFT

Time domain periodic convolution is equivalent to frequency domain multiplication, thus,

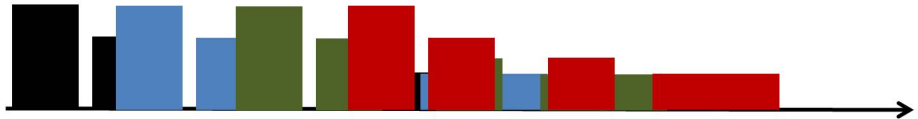
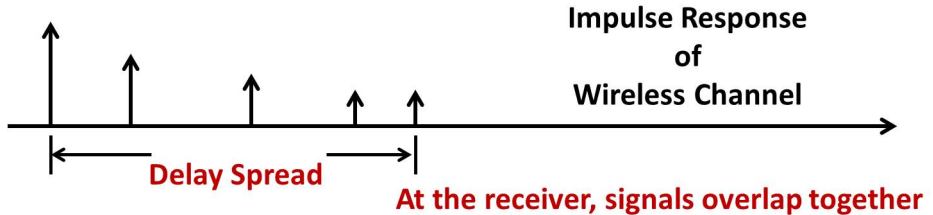
- $x[n] \circledast y[n] \longleftrightarrow N\tilde{X}[k]\tilde{Y}[k]$

Wireless Systems



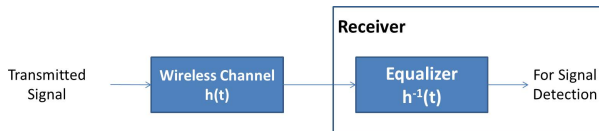
Delay spread causes inter-symbol interference

Inter-Symbol Interference (1/2)



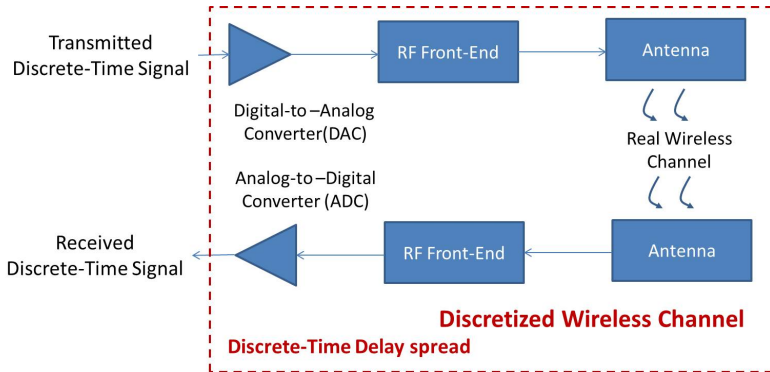
Inter-Symbol Interference (2/2)

- How to deliver signals without inter-symbol interference?
 - ▶ Suppose the duration of delay spread is ΔH seconds
- Approach 1: Send data on every ΔH seconds
- Approach 2: Channel equalizer



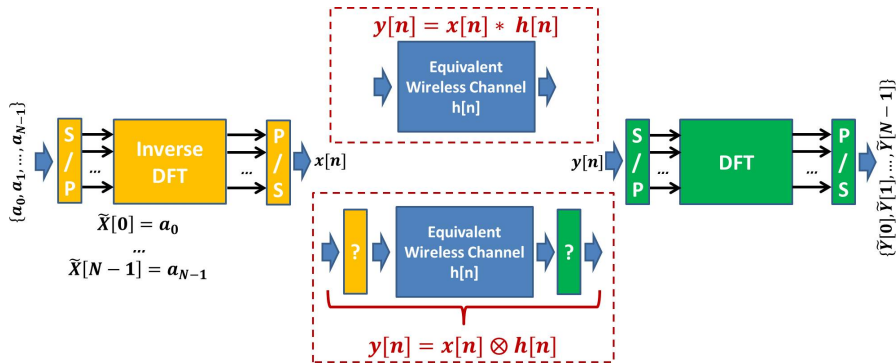
- Approach 3: Orthogonal Frequency Division Multiplexing
 - ▶ Pre-processing at the transmitter + post-processing at the receiver

Discretetize Wireless Channel



- Tx front-end + wireless channel + Rx front-end: approximately discrete-time LTI system
- Denote the impulse response as $h[n]$

OFDM at First Glance



- Signals are loaded in frequency domain
- Some mechanism is necessary to generate the effect of periodic convolution

OFDM Analysis

- Let $\{a_0, a_1, \dots, a_{N-1}\}$ be the sequence of bits to be delivered from the transmitter to the receiver.
- At the transmitter
 - ▶ we let

$$\tilde{X}[0] = a_0, \tilde{X}[1] = a_1, \dots, \tilde{X}[N-1] = a_{N-1}$$

- ▶ Take inverse DFT:

$$\{x[0], x[1], \dots, x[N-1]\} = DFT^{-1}\{\tilde{X}[0], \tilde{X}[1], \dots, \tilde{X}[N-1]\}$$

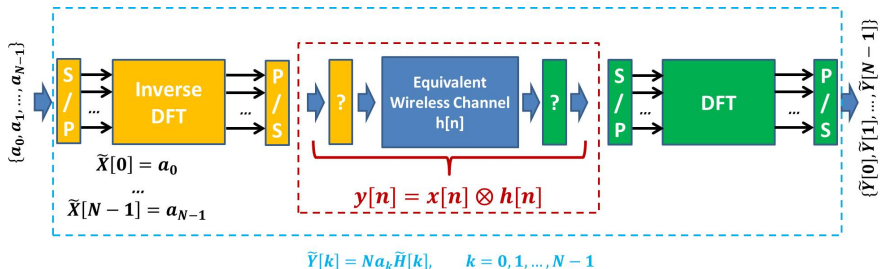
- With certain mechanism, the received signal becomes

$$y[n] = x[n] \circledast h[n]$$

- At the receiver, take DFT on $y[n]$:

$$\tilde{Y}[k] = N\tilde{X}[k]\tilde{H}[k] = Na_k\tilde{H}[k] \quad k = 0, 1, \dots, N-1$$

So, ...



- How to detect $\{a_k | \forall k\}$ from $\{\tilde{Y}[k] | \forall k\}$?
- How to design the blocks "?" ?

Channel Estimation & Signal Detection

- **Channel estimation:** Estimate the channel gain $\tilde{H}[k]$ ($k = 0, 1, \dots, N - 1$)
 - ▶ The transmitter sends the common information to the receiver $\{a_0^c, a_1^c, \dots, a_{N-1}^c\}$:

$$\tilde{Y}^c[k] = Na_k^c \tilde{H}[k] \quad k = 0, 1, \dots, N - 1$$

- ▶ Evaluate channel via

$$\tilde{H}[k] = \tilde{Y}^c[k] / (Na_k^c) \quad k = 0, 1, \dots, N - 1$$

- **Signal detection:** Detect the transmitter's signal via the knowledge of $\tilde{H}[k]$ ($k = 0, 1, \dots, N - 1$):
 - ▶ The transmitter sends the information to the receiver $\{a_0, a_1, \dots, a_{N-1}\}$:

$$\tilde{Y}[k] = Na_k \tilde{H}[k] \quad k = 0, 1, \dots, N - 1$$

- ▶ Detect signal via

$$a_k = \tilde{Y}[k] / (N\tilde{H}[k]) \quad k = 0, 1, \dots, N - 1$$

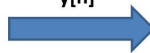
Convolution vs. Periodic Convolution

Transmission Signal
 $x[n]$

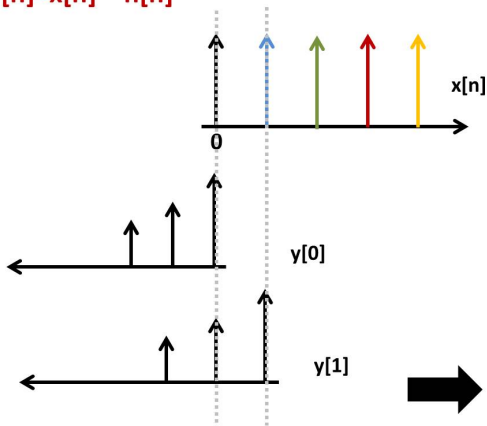
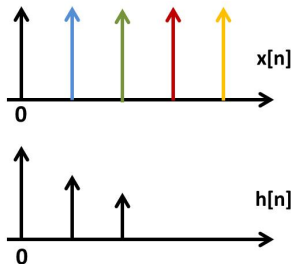


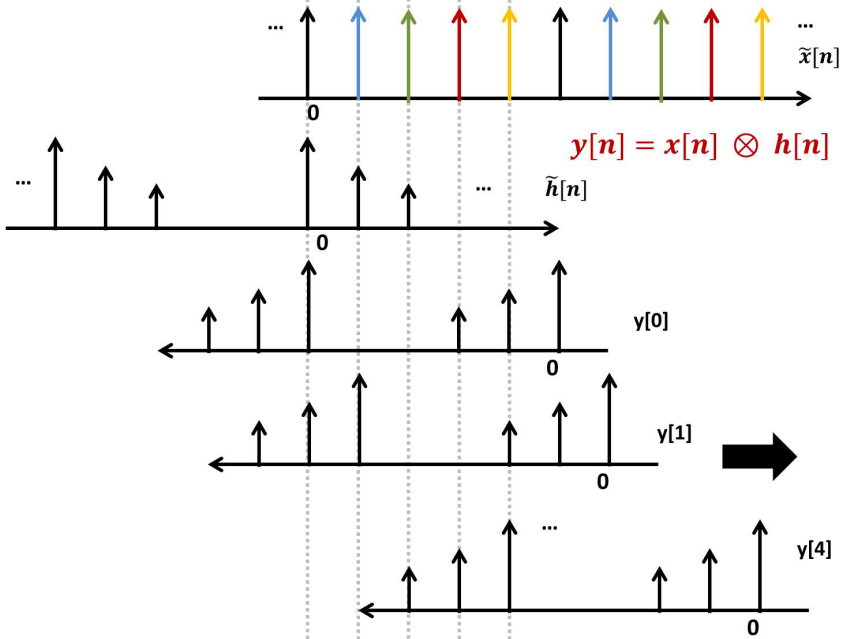
Equivalent Wireless Channel
DT LTI System $h[n]$

Received Signal
 $y[n]$



$$y[n] = x[n] * h[n]$$

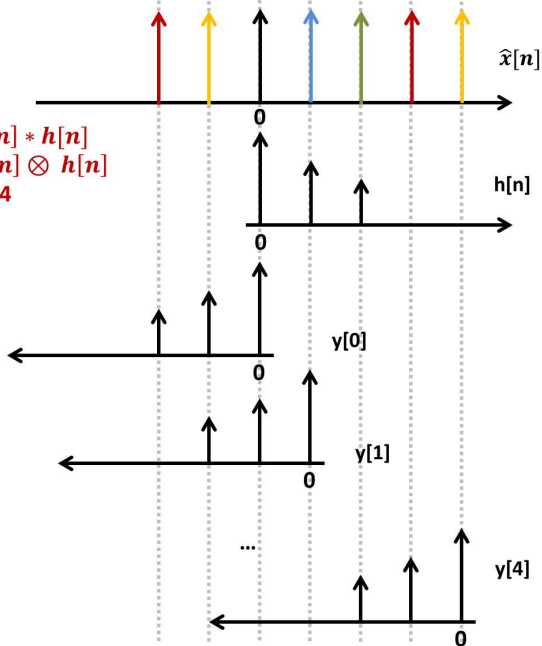




$$y[n] = \hat{x}[n] * h[n]$$

$$= x[n] \otimes h[n]$$

$$n=0,1,2,3,4$$



Observations

- Periodic convolution can be different from convolution
- In order to guarantee that

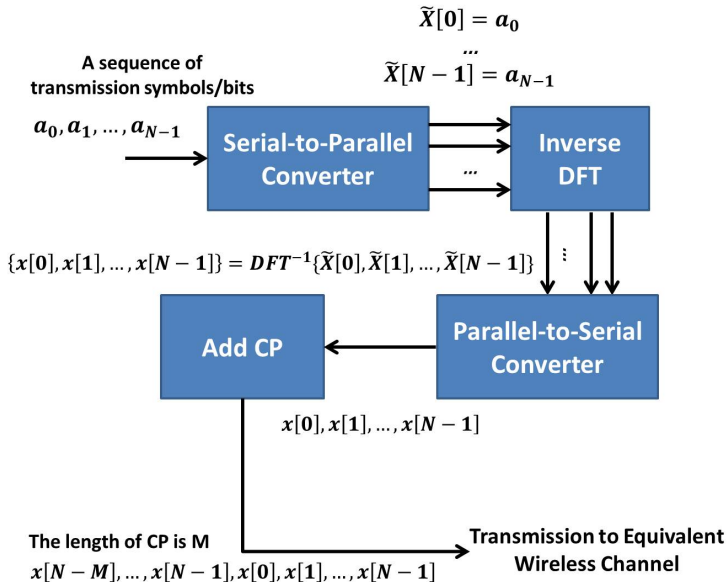
$$y[n] = x[n] \circledast h[n], \text{ for } n = 0, 1, 2, \dots, N - 1,$$

we should copy the last few samples of $x[n]$ to the beginning, which is named as **cyclic prefix**

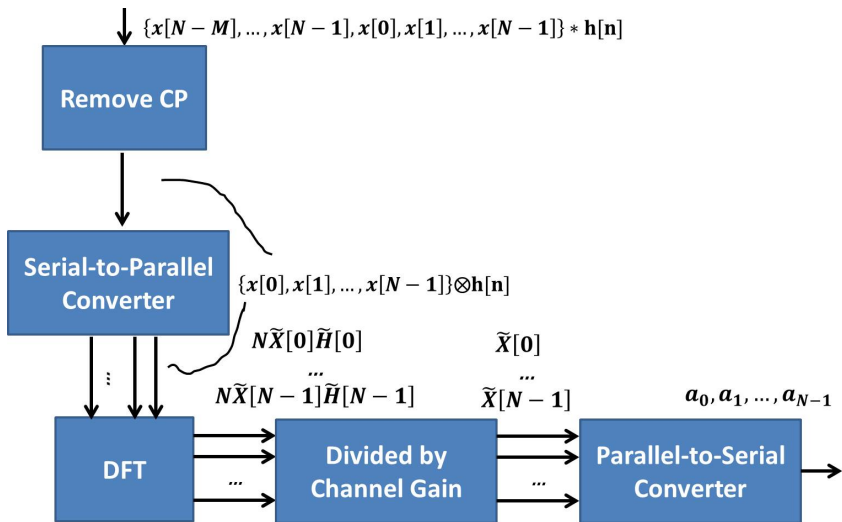
- ▶ How many samples should we copy?



Transmitter Structure of OFDM



Receiver Structure of OFDM



Reference & Tutorial

- Reference

- ▶ www.gaussianwaves.com/2011/05/introduction-to-ofdm-orthogonal-frequency-division-multiplexing-2/
- ▶ www.wirelesscommunication.nl/reference/chaptr05/ofdm/ofdmmath.htm
- ▶ home.deib.polimi.it/spalvier/sistemi_di_comunicazione/integrazione_lezioni/ofdm_tutorial.htm

- Tutorial Problems

- ▶ Textbook: 5.53, 5.54

